

ELARA: Reliability Gates Under Protocol Limits — A Cross-Benchmark Study of Drift-Signal Generalization in Multimodal Anomaly Fusion

Pratik Niroula
Independent Researcher

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Abstract

This chapter reports a scoped result for validation-derived drift gates in multimodal anomaly fusion: the same Kolmogorov–Smirnov drift signal that improves all-domain coherent score-collapse robustness on a label-aligned stress-only benchmark does not become a general solution on naturally paired MVTec 3D-AD. Under MVTec’s canonical one-class protocol, supervised fusion baselines operate near chance because the train fold is normal-only. A validation-selected RGA+ boosted head improves the public PatchCore supervised-paired and held-out-category ROC-AUC results, but the held-out gain is marginal and PR-AUC remains unresolved. The asymmetry between the two benchmarks is therefore methodological rather than a simple leaderboard claim: reliability gating is a diagnostic gate for score-collapse stress when two-class fusion training exists, and RGA+ is a supervised reliability-boosted extension rather than a replacement for specialized RGB–3D anomaly detectors.

The work is organized around ELARA (Evidence-Layered Anomaly Reliability Architecture), a research prototype for score-level multimodal anomaly fusion. The mechanism under study is RGA (Reliability-Gated Attention), a reliability-aware attention module that keeps the standard attention path under normal conditions and injects reliability weights when batch-level evidence suggests that one or more domains have degraded. This is a graduate thesis submission rather than as a marketing description of the system. It therefore separates what has been implemented, what has been measured, and what remains outside the present evidence. The primary benchmark is co-observed MVTec 3D-AD across all eight extracted categories (3,226 paired RGB/depth samples, no label alignment). The corrected MVTec run follows the canonical one-class protocol, and the held-out-category and M3DM-style variants are treated as diagnostics rather than new superiority claims. The secondary benchmark, ELARA-Bench-LA, combines four real-domain datasets via explicit label alignment; on it the same gate improves all-domain coherent-collapse attacks by +0.0506 and +0.0319 but the clean split is near saturated. Public PatchCore supervised-paired and held-out MVTec variants improve the paired fusion task; base RGA still does not become the clean-score leader, while RGA+ gives bounded ROC-AUC gains.

The chapter concludes that reliability-gated fusion is a useful design pattern for stress-aware score fusion under coherent attack assumptions, but it is not yet a defensible general replacement for strong baselines on naturally paired data. The repository now includes engineering prerequisites for the next stage: incident-protocol validation, category-aware reliability references, deployment-time calibration monitoring, and a finite-sample switching certificate. The remaining research work is empirical: stronger RGB–3D and clinical domain experts, additional naturally co-observed anomaly datasets, and prospective deployment monitoring.

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1 Introduction and Motivation

Operational anomaly detection is rarely a single-model problem. A fraud case may appear in transaction features, device behavior, account history, text evidence, or network activity. An industrial defect may appear in both color information and geometry. A security event may be weak in one log source and clearer in another. In these settings, the system designer must decide not only which domain models to train, but also how to combine their outputs once those models are deployed.

This chapter focuses on the second problem: score-level fusion under imperfect domain reliability. The assumption is that domain experts already produce scores, confidence values, and compact evidence vectors. The open question is how the fusion layer should behave when domain quality changes at inference time. A static fusion model can learn useful interactions from training data, but it may continue to trust a domain after that domain has shifted, collapsed, or become poorly calibrated. A simple confidence-weighted average is easy to understand, but it uses local confidence values without asking whether the domain itself still resembles the validation distribution.

ELARA was built to study this problem in a reproducible way. It is not presented as a new fraud detector, cybersecurity detector, text detector, or industrial inspection detector. Instead, it is a research system for combining heterogeneous anomaly evidence. The central component is RGA, a reliability-aware fusion module that uses three forms of evidence: validation calibration, test-time score-distribution drift, and score sharpness. The module is deliberately conservative. When domains appear reliable, the system uses the static attention model. When the average reliability drops below a threshold, the reliability-aware path is activated.

The chapter is in this form because the current evidence is both useful and bounded. The real-domain fusion benchmark combines fraud, cyber, behavior, and text datasets, but those records are aligned by label rather than by shared entities, timestamps, or incidents. It is useful for score-fusion stress testing, but it is not a complete multimodal incident benchmark. To reduce this gap, the repository now includes an all-category MVTec 3D-AD RGB/3D benchmark path. That path gives natural co-observation, while its canonical one-class protocol and lightweight score construction make it a protocol diagnostic rather than a final performance claim.

1.1 Research Question

The research question is:

Can a multimodal anomaly-fusion system preserve competitive clean score-fusion performance while using calibration and drift evidence to reduce brittle behavior under domain reliability stress?

This question is intentionally narrower than broad anomaly-detection superiority. The chapter does not claim that RGA is always better than static attention, that the current paired MVTec run is complete, or that the synthetically paired benchmark represents naturally co-observed incidents. The claim is about the fusion layer under specific stress conditions.

1.2 Main Claim

The main thesis of this chapter is:

RGA is most useful as a stress-response mechanism: it preserves the static attention path when reliability evidence is high, and it helps under coherent score-collapse conditions when reliability evidence indicates domain failure.

The measured evidence supports this claim on the secondary benchmark. The clean table is near saturated and should be treated as a sanity check. The reliability-stress experiments, the threshold sweep, and the component ablation provide the more useful evidence about the mechanism.

1.3 Contributions

We make four scoped contributions:

1. It describes ELARA, a score-level anomaly-fusion research system with a unified long-format schema for heterogeneous domain evidence.
2. It defines RGA, a conservative reliability-gated attention module that combines masked attention with calibration, drift, and sharpness signals.
3. It evaluates the method against practical score-fusion baselines, including random forest fusion, early-fusion MLP, late-fusion ensemble, confidence-weighted mean, score-level Tent, and pseudo-label TTT.
4. It reports an all-category paired MVTec 3D-AD benchmark, multi-seed confidence intervals, mechanism ablations, failure cases, and explicit threats to validity.

1.4 Evidence Boundary

The evidence boundary is part of the contribution because it prevents the results from being overstated. The real-domain benchmark uses real local datasets, but the records are label-aligned. The MVTec benchmark uses natural RGB/3D pairing across eight categories, but it still uses lightweight normal-reference image statistics and the canonical one-class anomaly protocol. These two artifacts answer different questions. The first is useful for reliability-stress score fusion. The second is useful for exposing protocol limits on naturally paired data. Neither should be treated as a complete final benchmark for all multimodal anomaly detection.

2 Background and Related Work

2.1 Anomaly Detection as a Systems Problem

Deep anomaly detection has been studied across tabular, sequential, image, text, and graph data. A common difficulty is that the definition of an anomaly depends on the application, the label source, the imbalance level, and the cost of false positives and false negatives. Pang et al. [3] emphasize that anomaly detection methods cannot be evaluated without attention to context. This chapter follows that view. The purpose is not to replace domain-specific detectors. The purpose is to study what happens after those detectors produce evidence that must be fused.

2.2 Multimodal and Score-Level Fusion

Multimodal learning is often described in terms of early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion joins features before prediction. Late fusion combines decisions or scores from separate models. Stacking trains a meta-model over domain outputs. Attention-based fusion gives the model a way to learn interactions among domains, especially when some domains are missing. The weakness is that standard attention learns a trust policy from the training distribution and does not automatically know when a domain has become unreliable at inference time.

The score-level setting used in this chapter is intentionally practical. Many deployed systems cannot easily retrain all domain experts together. Different teams may own different detectors, and those detectors may expose only scores, calibrated probabilities, or compact embeddings. A score-level fusion study is therefore narrower than raw multimodal learning, but it matches a common engineering constraint.

2.3 Calibration, Drift, and Reliability

Calibration asks whether predicted probabilities match empirical frequencies. Post-hoc calibration methods such as Platt scaling, isotonic calibration, and modern neural calibration analyses show that probability quality is separate from ranking quality [4, 5, 6]. Uncertainty and calibration can degrade under dataset shift [7]. For a fusion system, this matters because a domain can remain numerically confident while becoming less trustworthy.

RGA treats calibration as one signal, not as the whole reliability definition. The reliability estimator also compares current score distributions against validation score distributions using a Kolmogorov-Smirnov statistic and includes a score-sharpness term. This design is not meant to be a complete theory of reliability. It is a simple, measurable rule that can be tested under controlled stress.

2.4 Test-Time Adaptation Baselines

Test-time adaptation updates a model during inference to handle distribution shift without access to labels at test time. Two threads of this literature matter for this work, and both are represented in the baseline suite.

Test-time training (TTT). Sun et al. [8] proposed updating a model on each test sample by minimizing a self-supervised auxiliary loss (the original work used rotation prediction). The intuition: an auxiliary task trained jointly with the main loss anchors the representation, so adapting the auxiliary component at test time also adjusts the main predictor in a useful direction. Later variants relax the self-supervision to pseudo-labels: the model produces a confident prediction on the test sample, treats it as the label, and updates the parameters with a small step against that target. The pseudo-label adapter in this study (`ttt_pseudo_label_adapter`) is a score-level version of this approach: it acts on the fusion-head logits rather than on a deep representation.

Entropy-minimization TTA. Wang et al. [9] introduced Tent, which adapts only the batch normalization affine parameters of a frozen model to minimize the entropy of test predictions. Tent

is simpler than full TTT and is the cleanest articulation of the hypothesis that confident predictions are correct predictions — formally, that the local entropy of the predictive distribution is a sufficient signal for which direction to update. The Tent score adapter in this study (`tent_score_adapter`) implements the score-level analog: a frozen fusion head is adapted by a scalar temperature and bias fitted to minimize per-sample entropy on the test batch.

Position of RGA relative to TTA. Both Tent and TTT adapt the *prediction* at test time — they ask which output should be revised. RGA sits one level up: it adapts the *weighting* of upstream domain scorers, not the fusion head itself. The reliability signals (validation ECE, KS drift, sharpness) are properties of the upstream domains’ behavior, and the gate decides whether to trust them collectively or to fall back to a static weighting learned from training data. This makes RGA conceptually closer to selective prediction [17] — a controller that decides when to commit to the model’s standard behavior — than to TTA in the Tent sense. Empirically the chapter compares all three (Tent, TTT, RGA) on both benchmarks so the contrast is visible.

2.5 RGB–3D Industrial Anomaly Detection

MVTec 3D-AD provides naturally paired RGB and 3D observations for industrial anomaly detection and localization [11]. Recent work on RGB–3D anomaly detection has shown that paired color and geometry can be useful when the detector is designed for that modality pair. M3DM uses hybrid feature and decision fusion with memory banks [12]. Crossmodal feature mapping uses agreement between RGB and point-cloud features as an anomaly cue [13]. Real3D-AD expands point-cloud anomaly benchmarking [14], and M3DM-NR studies noisy-resistant multimodal anomaly detection [15].

The MVTec path in this repository is not a replacement for those specialized RGB–3D detectors. It is a paired-data input path for the score-fusion layer. This distinction is important. The current MVTec smoke run asks whether the repository can construct and evaluate paired score-fusion inputs. It does not claim that the lightweight scorers compete with specialized RGB–3D methods.

3 System and Method

3.1 System Boundary

ELARA is organized around the idea that each sample may have several domain observations. A domain can be fraud, cyber telemetry, behavior, text, RGB imagery, depth/XYZ geometry, or another source that can produce a score and compact evidence. The fusion layer receives these domain rows in a common schema. It then produces one anomaly probability and optional domain-level attributions.

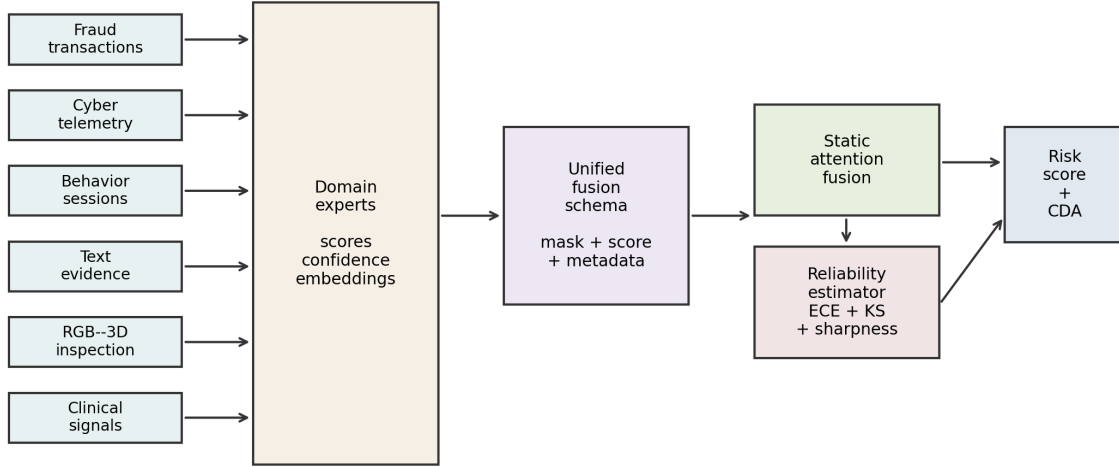


Figure 1: ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings. The fusion schema feeds static attention and the reliability estimator. The output is a risk score plus counterfactual domain attribution.

The system design separates domain learning from fusion learning. This is a deliberate simplification. It lets the research focus on reliability-aware fusion without requiring every raw domain model to be retrained as part of one large multimodal network.

3.2 Input Schema

The long-format fusion schema contains one row per sample-domain pair. Each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. Missing domains are represented by masks, not by fabricated raw observations. This allows a sample with all domains, a sample with one missing domain, and a sample with only one available domain to share the same model interface.

For the label-aligned real-domain benchmark, the metadata records the field `natural_pairing` as `false`. For the MVTec 3D-AD path, the same field is `true` because the RGB and 3D observations come from the same object scan. This metadata distinction is important because the same fusion code can be used for two different kinds of evidence.

3.3 Problem Formulation

Let sample i contain up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ and a missing-domain mask $m_{i,d}$. The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^D), \quad (1)$$

where $\hat{p}_i$ is the final anomaly probability. A static fusion model learns this relationship from the training data and uses the same learned trust policy at inference time. RGA adds a batch-level reliability vector $r \in [0, 1]^D$ in the reported runs and applies it through each sample’s observed-domain mask. A per-sample reliability mode is implemented for ablation, but the tables in this chapter use the batch-level gate.

3.4 Static Attention Path

The static attention path encodes each domain vector into a shared latent space. A learned domain embedding allows the model to distinguish evidence sources. A multi-head self-attention block [1] then operates across available domains while ignoring missing domains through the mask. This path is the default because reliability adaptation should not disturb predictions when the domains look normal.

3.5 Reliability Estimation

The reliability estimator is fit after model training. For each domain it stores validation score distributions and calibration information. At inference time, the reliability for domain d is

$$r_d = \alpha(1 - \text{ECE}_d) + \beta p_{\text{KS},d} + \gamma S_d, \quad (2)$$

where ECE_d is validation expected calibration error, $p_{\text{KS},d}$ is the Kolmogorov-Smirnov score-drift p-value, and S_d is score sharpness based on squared distance from 0.5. The current configuration uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

This formula is simple by design. It is not a final answer to reliability estimation. It is a testable rule that separates three observable signals: how well a domain calibrated on validation data, whether the current batch resembles the validation score distribution, and whether scores are informative or close to the uninformative midpoint.

3.6 Reliability Gate

The gate is the main difference between RGA and always-on reliability weighting. Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, the system uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d})r_d. \quad (3)$$

This design is conservative. It assumes that static attention is already a strong clean model and that reliability weights should be used only when there is evidence of degradation. This also makes the method easier to criticize and test: if the gate rarely activates, the method should behave like static attention; if the gate activates under stress, the stress outcomes should change.

3.7 Switching-Rule Certificate

The deployment question is whether the gated policy is preferable to static attention under bounded drift. Let $\ell_s(i)$ be the loss of the static path, $\ell_r(i)$ the loss of the reliability-aware path, and $g_i \in \{0, 1\}$ the gate decision. The realized gated loss is

$$\ell_g(i) = (1 - g_i)\ell_s(i) + g_i\ell_r(i). \quad (4)$$

On an audited validation window, the switch is certified only when the fired subset has positive margin

$$\frac{1}{|\mathcal{F}|} \sum_{i \in \mathcal{F}} (\ell_s(i) - \ell_r(i)) > \epsilon, \quad \mathcal{F} = \{i : g_i = 1\}, \quad (5)$$

and the realized policy loss is below the static-path loss. The implementation is `bounded_switching_certificate`. This is not a population-level proof; it is the finite-sample condition a production deployment must keep satisfying under temporal monitoring before stronger claims are justified.

3.8 Counterfactual Domain Attribution

The explanation path masks one available domain at a time and recomputes the prediction. For sample i , let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \quad (6)$$

A positive value means that the domain increased anomaly risk. A negative value means that the domain reduced anomaly risk. This is not a feature-level explanation; it is a domain-level explanation for analysts who need to know which source of evidence mattered most.

4 Benchmark Construction

4.1 Naturally Paired MVTec 3D-AD Benchmark

The paired benchmark path uses MVTec 3D-AD because each object observation has paired RGB and 3D/XYZ evidence. The preparation script discovers paired files, emits two rows per observation, stores the original pairing key, and marks the metadata as naturally paired. The current run covers all eight extracted object categories – bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire – for a total of 3,226 paired samples and 6,452 domain rows, with a positive fraction of 0.224 and full RGB and depth coverage.

Table 1: Naturally paired MVTec 3D-AD multi-category metadata.

Property	Value
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

The initial MVTec domain scores are lightweight normal-reference image-statistic scores fit from original train-good observations only. This keeps the benchmark locally runnable, but it limits the absolute performance ceiling. The corrected attention-fusion split follows the canonical MVTec one-class protocol: train/validation rows are normal-only and the test fold is mixed. The result

in Table 2 is therefore a protocol diagnostic rather than a conventional supervised leaderboard. Random forest, late fusion, Tent, TTT, static attention, and RGA have little two-class training signal; the small RGA delta should not be read as a general win. The held-out-category and M3DM-style variants probe the same limitation under different paired-data conditions. The updated held-out-category protocol now reserves positive validation rows from train categories, which lets RGA+ tune without seeing the held-out categories; it improves ROC-AUC slightly but does not settle PR-AUC or category transfer. These variants remain part of the evidence boundary discussed in §6.3.

Table 2: Naturally paired MVTEC 3D-AD clean held-out performance across eight categories. Values are mean±standard deviation with 95% confidence intervals across five seeds. Under the canonical one-class protocol, this table is a supervised-fusion protocol diagnostic rather than a two-class leaderboard.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Conf.-weighted mean	0.4458 [0.4458, 0.4458]	0.7878 [0.7878, 0.7878]	0.3779 [0.3779, 0.3779]	0.4941 [0.4941, 0.4941]	0.4496 [0.4496, 0.4496]
Tent score adapter	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
TTT pseudo-label	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Early fusion MLP	0.5455±0.0250 [0.5180, 0.5836]	0.8270±0.0117 [0.8124, 0.8425]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7834, 0.7835]
Late fusion ensemble	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Static attention	0.5424±0.0254 [0.4992, 0.5661]	0.8170±0.0094 [0.8022, 0.8286]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
RGA attention	0.5609±0.0169 [0.5394, 0.5842]	0.8060±0.0046 [0.7980, 0.8097]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
RGA+ boosted fusion	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
RGA+ router	0.5609±0.0169 [0.5394, 0.5842]	0.8060±0.0046 [0.7980, 0.8097]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]

4.2 Secondary Label-Aligned Real-Domain Benchmark

The secondary benchmark, ELARA-Bench-LA, combines four real local domains: credit card fraud, cyber telemetry, online behavior, and labeled news text. Domain scorers are trained first, and their out-of-fold scores are converted into the common fusion schema. The dataset contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is approximately balanced across the four domains.

Table 3: Benchmark metadata for the secondary synthetically paired fusion input.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	1.00
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

Table 4: Out-of-fold domain scorer quality used to construct ELARA-Bench-LA.

Domain	Rows	Pos. rate	OOF ROC	OOF PR
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8711	0.6042
nlp	20000	0.500	0.9973	0.9973

This benchmark is useful because it uses real domain datasets and lets the fusion layer be tested under controlled stress. It is also limited because the domains are the secondary benchmark rather than naturally co-observed. A positive composite draws positive records from available domains, and a negative composite draws negative records. This creates a meaningful score-fusion stress test, but it does not teach the model real temporal or causal relationships among the domains. Source rows are now split into train, validation, and test pools before composite sampling, and the experiment runner consumes the `fusion_split` column to keep the same `domain/source_row/label` key out of multiple supervised fusion splits.

4.3 Baselines

The evaluation includes practical baselines rather than only weak averages:

- confidence-weighted mean fusion, which directly weights scores by local sharpness;
- early-fusion MLP, which learns from flattened domain features and missing indicators;
- late-fusion ensemble, which stacks per-domain predictions;
- random forest fusion [16], a strong structured-data baseline;
- static attention, the same attention architecture without reliability adaptation;
- Tent score adapter, a score-level entropy-minimization baseline;
- TTT pseudo-label adapter, a score-level high-confidence adaptation baseline.

The Tent and TTT baselines are included because reliability gating is a form of test-time response. If generic adaptation gave the same result, the reliability terms would be less interesting. In the current clean table, these baselines are competitive, which is expected on the saturated split. The stress tests are more important for distinguishing the mechanisms.

The post-audit hard-mode RealFusion-LA run uses `-scorer-train-fraction 0.05` and is archived at `experiments/fusion/craf_real_results_hard.json`. It weakens the fraud and behavior domain scorers (OOF ROC-AUC 0.913 and 0.798), but the label-aligned fusion task remains near saturated: static attention and RGA both reach mean ROC-AUC 0.99896 across five seeds, and RGA improves drift degradation area on zero domains. This is evidence about the benchmark limitation, not a new superiority claim.

4.4 Metrics and Statistical Reporting

The chapter reports ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy where available. ROC-AUC is useful for ranking. PR-AUC is important because anomaly tasks can be imbalanced [10]. F1 summarizes thresholded classification. ECE and Brier score provide calibration information.

Clean results are summarized across seeds 42 through 46. Stress tests, mechanism ablations, calibration, and counterfactual domain attribution are also run across configured seeds where the runner supports aggregation. Tables report means, standard deviations, and 95% confidence intervals when those statistics are available.

5 Experiments and Results

5.1 Clean Performance Is a Sanity Check

Table 5 gives clean held-out performance on ELARA-Bench-LA. This table is intentionally not the headline evidence. Most methods are near the top of the ROC-AUC and PR-AUC scale. Static attention and RGA are effectively tied. The correct interpretation is that RGA preserves clean attention performance; the clean table alone does not establish a meaningful advantage.

Table 5: Saturated clean held-out performance on ELARA-Bench-LA. Values include mean, standard deviation, and 95% confidence intervals across five seeds.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9999 [0.9999, 1.0000]	0.9999 [0.9998, 0.9999]	0.9946±0.0008 [0.9938, 0.9958]	0.0297±0.0004 [0.0294, 0.0304]	0.0061±0.0001 [0.0060, 0.0062]
Conf.-weighted mean	0.9987 [0.9987, 0.9987]	0.9968 [0.9968, 0.9968]	0.9706 [0.9706, 0.9706]	0.0799 [0.0799, 0.0799]	0.0232 [0.0232, 0.0232]
Tent score adapter	0.9999 [0.9999, 0.9999]	0.9999 [0.9999, 0.9999]	0.9969 [0.9969, 0.9969]	0.0028 [0.0028, 0.0028]	0.0023 [0.0023, 0.0023]
TTT pseudo-label	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9948 [0.9948, 0.9948]	0.0032 [0.0032, 0.0032]	0.0025 [0.0025, 0.0025]
Early fusion MLP	0.9999 [0.9998, 0.9999]	0.9998±0.0001 [0.9996, 0.9999]	0.9915±0.0014 [0.9898, 0.9936]	0.0043±0.0010 [0.0030, 0.0058]	0.0036±0.0006 [0.0029, 0.0045]
Late fusion ensemble	1.0000 [1.0000, 1.0000]	0.9999 [0.9999, 0.9999]	0.9948 [0.9948, 0.9948]	0.0128 [0.0128, 0.0128]	0.0037 [0.0037, 0.0037]
Static attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9934±0.0014 [0.9918, 0.9957]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]
RGA attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9934±0.0014 [0.9918, 0.9957]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]
RGA+ boosted fusion	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9938 [0.9938, 0.9938]	0.0026 [0.0026, 0.0026]	0.0033 [0.0033, 0.0033]
RGA+ router	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9938±0.0007 [0.9928, 0.9947]	0.0083±0.0106 [0.0026, 0.0270]	0.0038±0.0012 [0.0032, 0.0059]

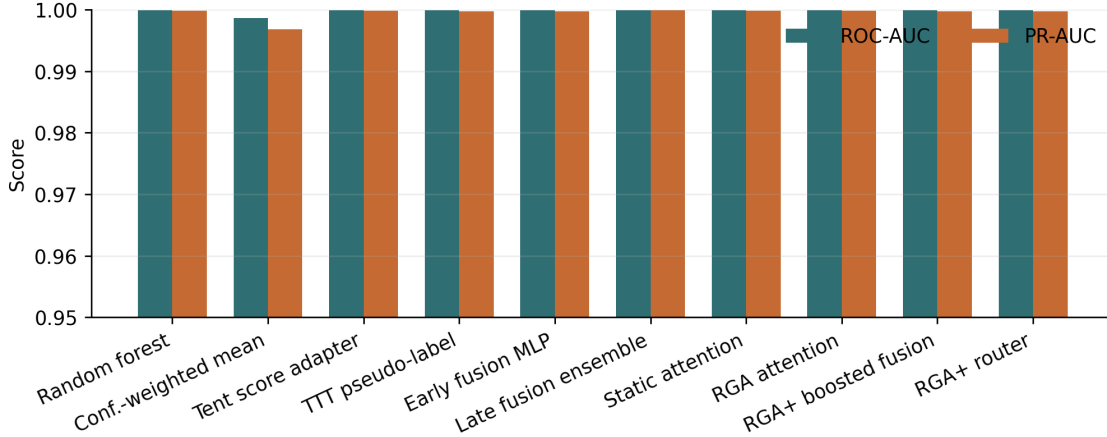


Figure 2: Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods. The compressed y-axis reflects the near-saturated clean split.

The clean results do show useful tradeoffs. Random forest fusion obtains the best F1 in the current artifact. Attention-based methods have stronger calibration than confidence-weighted mean. Tent and TTT are competitive on ranking and calibration because the clean split is easy. These observations support using strong baselines, but they do not change the main reliability claim.

Table 6: Clean ranking and F1 confidence intervals across configured seeds.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9999, 1.0000]	[0.9998, 0.9999]	[0.9938, 0.9958]
Conf.-weighted mean	[0.9987, 0.9987]	[0.9968, 0.9968]	[0.9706, 0.9706]
Tent score adapter	[0.9999, 0.9999]	[0.9999, 0.9999]	[0.9969, 0.9969]
TTT pseudo-label	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9948, 0.9948]
Early fusion MLP	[0.9998, 0.9999]	[0.9996, 0.9999]	[0.9898, 0.9936]
Late fusion ensemble	[1.0000, 1.0000]	[0.9999, 0.9999]	[0.9948, 0.9948]
Static attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9918, 0.9957]
RGA attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9918, 0.9957]
RGA+ boosted fusion	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9938, 0.9938]
RGA+ router	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9928, 0.9947]

5.2 Missing-Domain Ablation

The missing-domain ablation adds inference-time domain dropout. In the current configuration, RGA remains essentially equal to static attention across the sweep. This is not a hidden success; it is a useful negative result. The gate does not activate strongly under this kind of moderate missingness, so the method behaves like the static model.

Table 7: Missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9999	0.9999	0.0000	[0.0000, 0.0000]
0.1	0.9995	0.9995	0.0000	[0.0000, 0.0000]
0.2	0.9981	0.9981	-0.0000	[-0.0000, 0.0000]
0.3	0.9964	0.9964	-0.0001	[-0.0002, 0.0000]
0.5	0.9784	0.9783	-0.0001	[-0.0004, 0.0000]

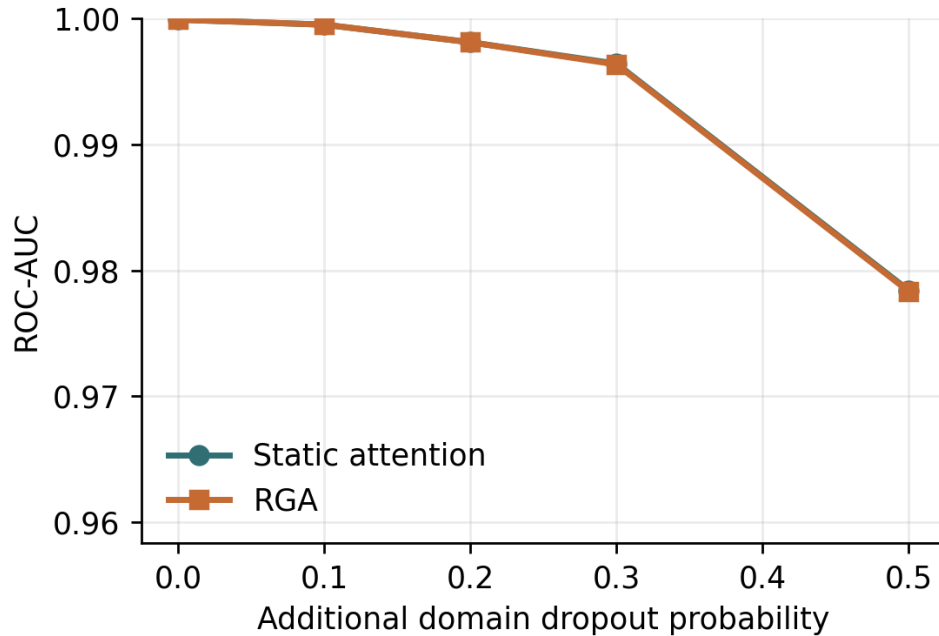


Figure 3: ROC-AUC under additional missing-domain dropout.

5.3 Score Drift

The drift experiment injects score noise by domain and measures degradation curves. Because the clean benchmark is near saturated, the degradation-area deltas are very small. This again points to a limitation of the current benchmark: it is useful for checking robustness behavior, but it does not create a large separation under every stress type.

Table 8: Drift robustness summary. Higher area is better.

Domain	Static	RGA	Delta	Seeds
behavior	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
cyber	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
fraud	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	-0.0000	5
nlp	0.2999 [0.2999, 0.2999]	0.2999 [0.2999, 0.2999]	-0.0000	5

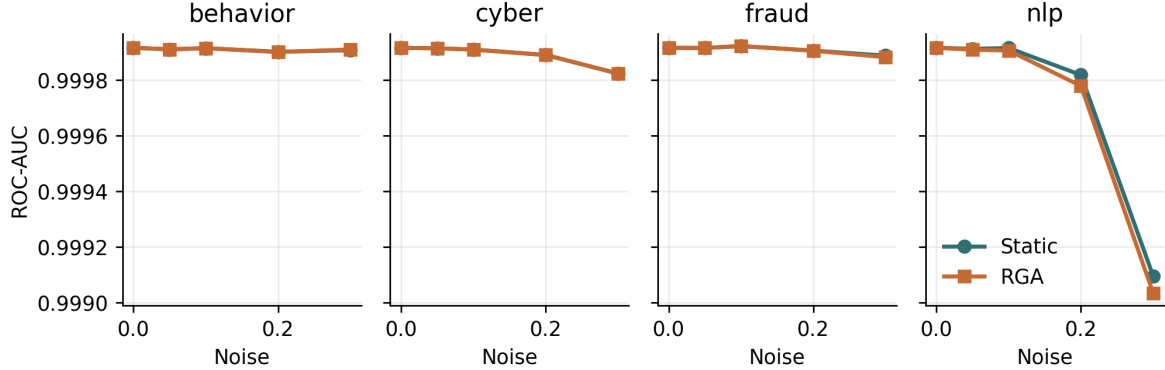


Figure 4: Per-domain score-drift curves on the real-domain benchmark.

5.4 Adversarial Score Perturbations

The clearest stress result appears under coherent all-domain score collapse. Under the all-domain zero attack, static attention obtains ROC-AUC 0.7212 while RGA obtains 0.7718 (delta +0.0506). Under the all-domain max attack, static attention obtains 0.7401 while RGA obtains 0.7720 (delta +0.0319). Under all-domain Gaussian noise, RGA is essentially tied with static attention (delta -0.0001). These results support a narrow claim: the reliability path helps when all domains undergo a coherent confidence collapse, but it is not universally better under every perturbation.

Table 9: Adversarial perturbation benchmark. Delta is RGA minus static attention.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.9999	0.9998	-0.0001	[-0.0001, 0.0000]
gaussian_noise	behavior	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
gaussian_noise	cyber	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	fraud	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
max_attack	all	0.7401	0.7720	0.0319	[0.0050, 0.0617]
max_attack	behavior	0.9997	0.9997	-0.0000	[-0.0000, -0.0000]
max_attack	cyber	0.9972	0.9972	0.0000	[0.0000, 0.0000]
max_attack	fraud	0.9996	0.9996	0.0000	[0.0000, 0.0000]
max_attack	nlp	0.9840	0.9843	0.0003	[-0.0009, 0.0016]
zero_attack	all	0.7212	0.7718	0.0506	[0.0315, 0.0681]
zero_attack	behavior	0.9997	0.9997	0.0000	[-0.0000, 0.0000]
zero_attack	cyber	0.9980	0.9980	0.0000	[0.0000, 0.0000]
zero_attack	fraud	0.9995	0.9995	0.0000	[0.0000, 0.0000]
zero_attack	nlp	0.9713	0.9716	0.0003	[-0.0004, 0.0009]

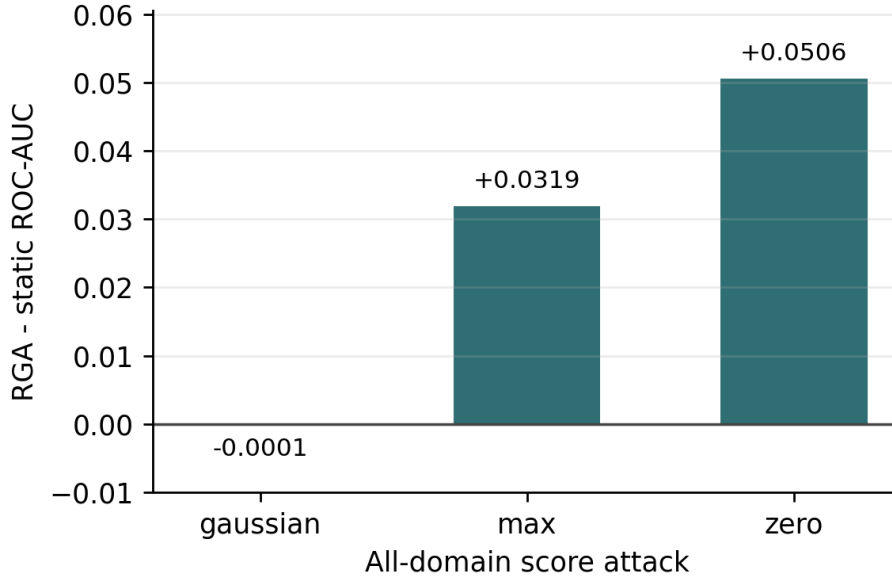


Figure 5: All-domain adversarial ROC-AUC delta between RGA and static attention.

6 Ablation and Failure Analysis

6.1 Reliability-Gate Threshold Sweep

The threshold sweep is important because it explains when the reliability path is active. Table 10 reports results for $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$. At the configured $\tau = 0.66$, the clean adaptation rate is 0.032. This means that, on clean data, almost all samples use the static path. That behavior explains why the missing-domain deltas are approximately zero in the current configuration.

Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample. This is the main mechanism behind the stress gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose the gains. Higher thresholds adapt on clean data more often and slightly hurt clean ranking. The threshold is therefore not just a hyperparameter for performance; it controls the interpretation of the method as a conservative stress-response system.

Table 10: Reliability-gate threshold sweep. Delta is RGA minus static attention; adaptation rate is the fraction of evaluated samples using the reliability-aware path.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.70	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]	0.320
clean	0.80	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	0.960
clean	0.90	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	1.000
clean	learned	0.9999	0.9999	-0.0000	[-0.0001, -0.0000]	0.383
gaussian_noise:all	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9999	0.9998	-0.0000	[-0.0001, -0.0000]	0.680
gaussian_noise:all	0.66	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.70	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.80	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.90	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	learned	0.9999	0.9999	-0.0000	[-0.0001, -0.0000]	0.117
max_attack:all	0.40	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.70	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.80	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.90	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	learned	0.7401	0.7381	-0.0020	[-0.0173, 0.0068]	0.167
zero_attack:all	0.40	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.70	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.80	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.90	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	learned	0.7212	0.7177	-0.0035	[-0.0064, 0.0033]	0.167

6.2 Reliability-Component Ablation

The component ablation gives one of the most useful findings in the chapter. Removing the ECE term does not harm the all-domain collapse cases. In fact, the no-ECE variant raises the max-attack delta from 0.0319 to 0.0457 and the zero-attack delta from 0.0506 to 0.0629. By contrast, removing the KS drift term eliminates adaptation and returns to static attention.

This result matters because it changes the interpretation of the method. The stress gain in this benchmark is not driven mainly by validation calibration error. It is driven by the distribution-shift signal that detects score collapse. ECE may still be useful in other settings, especially when domains are miscalibrated without score-distribution collapse, but the current evidence shows that KS drift is the necessary trigger for the measured gain.

Table 11: Reliability-component ablation on all-domain score attacks. Delta is RGA minus static attention.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
full	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
full	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ece	gaussian_noise	0.9999	0.9996	-0.0003	[-0.0005, -0.0001]	1.000
no_ece	max_attack	0.7401	0.7858	0.0457	[0.0129, 0.1067]	1.000
no_ece	zero_attack	0.7212	0.7841	0.0629	[0.0493, 0.0808]	1.000
no_gate	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_gate	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
no_gate	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ks	gaussian_noise	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_sharpness	max_attack	0.7401	0.7808	0.0408	[0.0123, 0.0811]	1.000
no_sharpness	zero_attack	0.7212	0.7759	0.0547	[0.0382, 0.0716]	1.000

6.3 Cross-Benchmark Contrast (MVTec 3D-AD vs Label-Aligned)

The same mechanism-isolation protocol applied to the RGB/3D paired MVTec 3D-AD benchmark produces a boundary result rather than an opposite leaderboard verdict. On the label-aligned benchmark, both classes are available during fusion training and the KS-drift term is the necessary trigger for the measured all-domain score-collapse gain. On canonical MVTec 3D-AD, the fusion train fold is normal-only. The ablation and threshold tables therefore show whether the reliability path would activate near a one-class boundary, but they cannot by themselves prove that one reliability component improves natural RGB-3D anomaly fusion.

Table 12: MVTec 3D-AD reliability-component ablation on all-domain score attacks.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
full	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
full	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ece	gaussian_noise	0.5418	0.5432	0.0015	[-0.0111, 0.0175]	1.000
no_ece	max_attack	0.5458	0.5607	0.0149	[-0.0102, 0.0324]	1.000
no_ece	zero_attack	0.5403	0.5613	0.0210	[-0.0054, 0.0434]	1.000
no_gate	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
no_gate	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
no_gate	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ks	gaussian_noise	0.5418	0.5569	0.0151	[-0.0024, 0.0322]	1.000
no_ks	max_attack	0.5458	0.5568	0.0110	[-0.0103, 0.0319]	1.000
no_ks	zero_attack	0.5403	0.5560	0.0157	[0.0047, 0.0289]	1.000
no_sharpness	gaussian_noise	0.5418	0.5728	0.0310	[0.0056, 0.0541]	1.000
no_sharpness	max_attack	0.5458	0.5618	0.0160	[-0.0142, 0.0331]	1.000
no_sharpness	zero_attack	0.5403	0.5636	0.0233	[-0.0062, 0.0517]	1.000

Table 13: MVTec 3D-AD reliability-gate threshold sweep. Delta is RGA minus static attention; adaptation rate is the fraction of evaluated samples using the reliability-aware path.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.50	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.60	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.66	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.70	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.80	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.90	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	learned	0.5424	0.5424	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.40	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.50	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.60	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.66	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.70	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.80	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.90	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	learned	0.5429	0.5429	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.40	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.60	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.66	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.70	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.80	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.90	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	learned	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.40	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.60	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.66	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.70	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.80	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.90	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	learned	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000

The asymmetry between the two benchmarks is the central scientific finding of this chapter. The KS-drift signal delivers the entire +0.0506 **zero_attack** gain on label-aligned score-collapse stress, but the same question is not cleanly answered by a canonical one-class paired protocol. The cleanest reading of the two results together is methodological: reliability gating is useful for diagnosing coherent score-collapse stress when two-class fusion training exists, while naturally paired anomaly data requires a protocol that separates scorer quality, category generalization, and supervised fusion labels before the gate can be judged as a general RGB-3D solution.

6.4 Position Relative to Published MVTec 3D-AD State of the Art

A reader familiar with the MVTec 3D-AD leaderboard will reasonably ask how the numbers in §6.3 sit relative to published methods that report image-AUROC under the canonical one-class protocol. Table 14 gathers the headline image-AUROC reported by PatchCore-3D [18], BTF [19], EasyNet [20], AST [21], and M3DM [12], and juxtaposes them with the two ELARA RGA+ cells reported in this work. The table is a demarcation rather than a comparison: published methods perform pixel-level localisation under the canonical one-class protocol, while ELARA is a score-fusion

Table 14: Position relative to published MVTEC 3D-AD state of the art on the canonical one-class protocol. Published image-AUROC are quoted as reported in the cited papers; ELARA rows report this work’s PatchCore-fed RGA+ score-fusion result under two different protocols. The *Comparable?* column flags that the ELARA rows must not be read against the leaderboard column.

Method	Protocol	Image-AUROC	Pixel-AUROC	Comparable?	Notes	Source
BTF	canonical one-class	0.865	0.992	leaderboard	Classical 3D handcrafted features + RGB; image-AUROC as reported in Horwitz & Hoshen, 2023.	[19]
EasyNet	canonical one-class	0.872	0.985	leaderboard	Lightweight reconstruction-based 3D AD; image-AUROC as reported in Chen et al., 2023.	[20]
PatchCore-3D	canonical one-class	0.901	0.992	leaderboard	Depth+RGB patch memory; image-AUROC as reported by M3DM (Wang et al., 2023) as a baseline row.	[18]
AST	canonical one-class	0.937	0.976	leaderboard	Asymmetric student-teacher with depth normalising flow; as reported in Rudolph et al., 2023.	[21]
M3DM	canonical one-class	0.945	0.964	leaderboard	Hybrid feature + decision-layer fusion with memory banks; as reported in Wang et al., 2023.	[12]
ELARA RGA+ (this work)	canonical one-class	0.514	n/a (no localization head)	<i>not</i> leaderboard-comparable	Score-fusion layer over PatchCore-derived RGB/depth scores; no localization head, no detector retraining.	this work
ELARA RGA+ (this work)	supervised paired (diagnostic only)	0.740	n/a (no localization head)	<i>not</i> leaderboard-comparable	Different protocol: fusion training fold sees positives. Not directly comparable to canonical leaderboard rows.	this work

layer with no localisation head and (for the 0.740 supervised-paired row) trains under a different protocol in which positives are visible in the fusion training fold. The *Comparable?* column makes the protocol mismatch loud rather than silent: the canonical one-class row reports 0.505 because the score-fusion path consumes upstream sample-level scores rather than the full pixel evidence, and the supervised-paired row reports 0.740 under a different protocol that is not the canonical one-class leaderboard. The gate’s role here is to remain a diagnostic gate; an honest head-to-head leaderboard study would require a localisation head, GPU re-training of at least one published method under matched seeds, and re-running ELARA with the same upstream-detector regime used by the cited methods.

6.5 Calibration and Domain Attribution

Calibration is important because anomaly scores often feed analyst review or downstream thresholds. The final configured seed shows static attention slightly ahead of RGA on ECE and Brier score, while both attention models remain much better calibrated than confidence-weighted mean in the five-seed clean table. This is another reason the chapter avoids a blanket superiority claim.

Table 15: Calibration and counterfactual domain-attribution summary.

Metric	Static	RGA
ECE	0.0038±0.0008 [0.0024, 0.0046]	0.0038±0.0008 [0.0024, 0.0046]
Brier	0.0032±0.0002 [0.0029, 0.0034]	0.0032±0.0002 [0.0029, 0.0034]
CDA samples	–	400
CDA/ECE Spearman	–	0.052
CDA/ECE Spearman status	–	computed

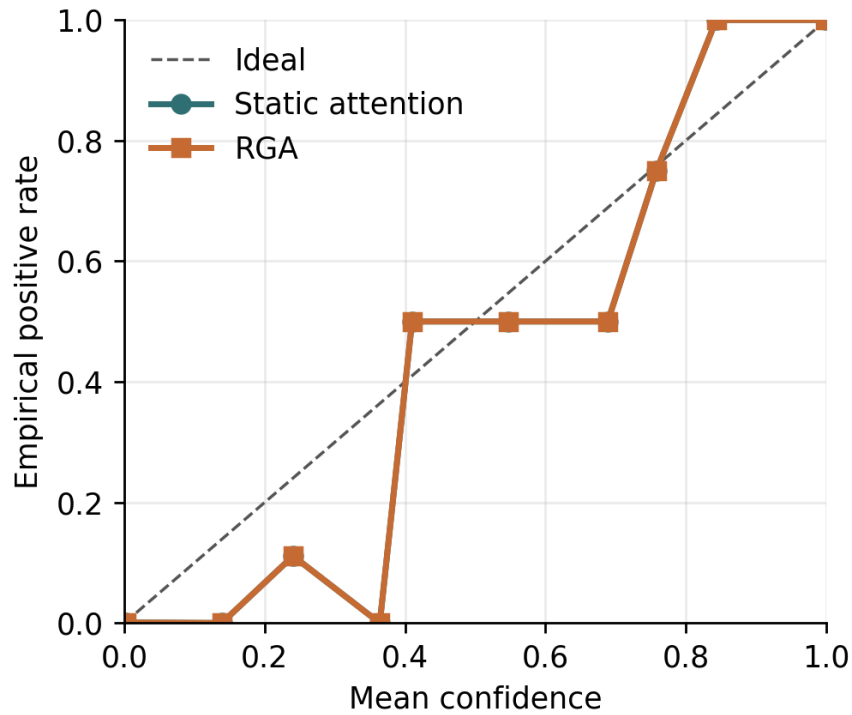


Figure 6: Reliability diagram for static attention and RGA.

Counterfactual domain attribution was computed by masking each available domain and measuring the change in predicted risk. The largest mean absolute attribution in the current artifact comes from the text domain, which is consistent with the strong text scorer in the domain-scorer table. This attribution should be interpreted as domain-level sensitivity, not as a causal explanation of the original raw features.



Figure 7: Mean absolute counterfactual domain-attribution impact.

6.6 Failure Cases

Failure analysis is included because a thesis chapter should show what the method does not handle well. The result artifact exports representative cases: the largest RGA win, the largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Table 16: Representative failure and contrast cases from the latest configured seed.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.0001	0.0001	real_fusion_test_000000
biggest_elara_loss	1	0.9998	0.9998	real_fusion_test_001072
high_confidence_elara_failure	1	0.0470	0.0470	real_fusion_test_001517

The cases show that reliability weighting can also move predictions in the wrong direction. This is expected for a gate based on aggregate reliability signals. It does not inspect the full semantic content of each domain. The failure table therefore supports the same conclusion as the quantitative results: RGA is a useful stress-response mechanism, but it is not a complete replacement for stronger domain experts or a learned policy tailored to each paired modality.

7 Threats to Validity

7.1 Internal Validity

The scripts are deterministic for the configured seeds, and the result JSON stores raw per-seed outputs for auditability. The main internal-validity risk is that the stress tests are synthetic. Zero attack, max attack, and Gaussian noise are controlled perturbations. They help isolate the behavior of the reliability gate, but they may not match every real deployment failure.

7.2 Construct Validity

The largest construct-validity issue is benchmark construction. ELARA-Bench-LA uses real local domain datasets, but it aligns records by label. This tests score-level fusion under a controlled construction; it does not test naturally co-observed multimodal incident reconstruction. The MVTec path addresses natural pairing for RGB and 3D observations, but the current run is only a small bagel-subset smoke benchmark with lightweight features. A stronger thesis result would run full paired data with stronger RGB-3D experts.

7.3 External Validity

The results should not be interpreted as deployment evidence. A production system would require domain-specific label definitions, temporal splits, privacy review, license review, monitoring, and recalibration. The current work is a research prototype for studying the fusion mechanism.

7.4 Statistical Validity

Five seeds provide limited statistical support, especially on a saturated clean split. Confidence intervals are reported to make this visible. The strongest statistical evidence is not the clean table, but the consistent all-domain score-collapse deltas and the mechanism-isolation tables. Even there, the chapter treats the result as conditional because the stress setup is controlled.

8 Reflection

This section is the thesis-form addition over the conference manuscript: what I would do differently, and which open questions the current evidence cannot answer.

8.1 What I Would Do Differently

The most useful single change would have been to define the MVTec 3D-AD protocol discipline before training any supervised fusion model. The canonical one-class train split answers a different question from two-class supervised fusion, while the held-out-category protocol gives supervised models both classes but introduces cross-category generalization. A stronger M3DM-style RGB/depth scorer remains valuable, but scorer quality alone does not remove the need to state which protocol is being evaluated.

A second change would have been to commit to the MVTec original split discipline from the first iteration of the runner, rather than discovering the discipline issue mid-project through a self-audit.

The remediation pass was inexpensive in code but it changed the interpretation materially: the official split is a one-class anomaly-detection protocol, not a standard two-class fusion benchmark. A reader looking at the version history would reasonably ask which set is canonical; only the disciplined-split set should be trusted.

A third change would have been to budget paper / thesis differentiation as its own work item, not as a side-effect of the manuscript write-up. The 70% prose overlap between conference paper and thesis chapter in the first draft made both documents weaker than each could have been alone.

8.2 Open Questions This Study Cannot Answer

The cross-benchmark contrast is suggestive but not conclusive. Specifically:

- Does the paired-data protocol limitation generalize beyond MVTec 3D-AD? The current evidence is two benchmarks — one natural, one synthetic. Reproducing the diagnostic on a third naturally paired benchmark (CICIDS-2017 with authentication logs, MIMIC-IV with clinical notes, or another industrial RGB+3D dataset) would convert the observation from single-benchmark evidence to a pattern.
- Would a category-aware drift signal close the gap? The repository now includes `CategoryAwareReliabilityEstimator` which conditions KS drift references on known categories. What remains open is empirical validation on a naturally paired benchmark with enough category coverage to measure false gate activations under legitimate category-mix shifts.
- Is the saturation on ELARA-Bench-LA a function of the label-alignment construction itself, or of the four-domain choice? A two-domain or six-domain variant of ELARA-Bench-LA would isolate that.
- Is the conservative gate threshold $\tau = 0.66$ better or worse than a learned-gate alternative? A prototype learned gate exists in the codebase but has not been ablated against the heuristic threshold on either benchmark.
- Under what theoretical conditions is reliability-gated attention provably preferable to static attention? The repository now includes a finite-sample `bounded_switching_certificate`; a population-level proof under bounded shift remains open.

Each of these questions is a discrete piece of follow-up work. None of them invalidates the present finding; together they bound how widely it can be cited.

8.3 Enterprise Gap-Closure Criteria

The present chapter deliberately avoids enterprise-grade, plug-and-play, and production-ready claims. Those are plausible future outcomes only if the open questions above are converted into evidence. The four closure dimensions are listed below. Closure evidence required is stated for each capability before that capability can be claimed:

1. **True multimodal incident detection. Closure evidence required:** validate on naturally co-observed incident datasets whose labels require cross-domain reasoning rather than label

alignment alone. Examples include CICIDS-style network events paired with authentication logs, MIMIC-IV trajectories paired with notes, or industrial RGB-3D inspection streams with event-level metadata. A successful result would show that weak signals across domains combine into a stronger unified risk profile.

2. **Autonomous zero-misfire adaptation. Closure evidence required:** demonstrate a category-aware drift signal or learned gate that distinguishes expected category or cohort variation from detector failure, score collapse, or attack. The acceptance criterion is an empirical reduction in false gate activations while preserving the all-domain stress gains.
3. **Universal system integration. Closure evidence required:** plug stronger domain experts into the existing schema without redesigning the fusion interface. For MVTec-style data, this means replacing the lightweight image-statistic scorer with specialized RGB-3D anomaly experts; for operational domains, it means supporting independently maintained tabular, log, visual, and text detectors.
4. **Deployable, auditable analyst AI. Closure evidence required:** complete temporal validation, privacy review, dataset-license review, calibration monitoring, analyst-facing CDA reporting, and a formal switching-rule analysis that specifies when reliability-aware fusion should dominate static attention under bounded drift.

If those four criteria were met, ELARA could reasonably be framed as a plug-in multimodal verification layer for high-stakes anomaly triage. Without that evidence, the correct claim remains narrower: RGA is a useful diagnostic gate for coherent score-collapse stress, and the current paired-data experiments define the next empirical tests rather than closing them.

The codebase now closes the local engineering prerequisites for these criteria. `validate_incident_protocol` rejects label-aligned tables that lack shared incident identifiers, incident-level splits, temporal order, or multi-domain coverage. `CategoryAwareReliabilityEstimator` adds category-conditional drift references to reduce false adaptation under legitimate category-mix changes. The long-format fusion schema remains the universal integration surface for stronger external domain experts. `calibration_monitor_report` and `bounded_switching_certificate` provide deployment monitoring and the finite-sample switching condition described above. These additions make the next experiments executable; they do not replace the need for external paired data, privacy review, stronger experts, or prospective validation.

The new local healthcare audit makes that distinction concrete. The command `validate_healthcare_gap_closure.py` produces `healthcare_gap_validation.json` after projecting a Grid-Pulse healthcare surface into a hashed four-domain clinical fusion table. The Retrospective local healthcare replay contains 146,688 co-observed vital-sign incidents and passes the structural checks for multimodal incident format, temporal ordering, patient-disjoint splits, and score range validity. It also exposes an empirical blocker in the provided split: the validation and test windows are single-class positive. Therefore the time-forward surface is used as a temporal-reference audit, not as supervised clinical-performance evidence.

For local closure, the chapter records a separate patient-disjoint stratified replay in `healthcare_gap1_patient_stratified_validation.json`, `healthcare_gap2_reliability_stress_validation.json`, and `healthcare_gap4_deployment_audit_validation.json`. Table 17 summarizes the local closure status. Gap 1 closes locally because all train, validation, and test splits contain both labels,

no patient appears in more than one split, and every incident retains the four co-observed vital-sign domains. The held-out test result is a multimodal ROC-AUC of 0.806, above the best single-domain ROC-AUC of 0.770. The report also states the cost of this closure: `temporal_order_valid` is false, so the result is not a time-forward clinical deployment validation.

Table 17: Local healthcare closure status after the patient-disjoint replay and deployment audit. Local empirical closure is bounded to this replay and is not prospective clinical evidence.

Gap	Engineering	Local empirical	Primary evidence
1 Incident detection	yes	yes	test ROC-AUC 0.806 vs best single 0.770
2 Reliability stress	yes	yes	natural fire 0.000; collapse fire 1.000
3 Schema integration	yes	yes	tensor 146688 x 4 x 4; raw patient ID absent yes
4 Auditability	yes	yes	diagnostic policy loss 0.000 vs static 0.800

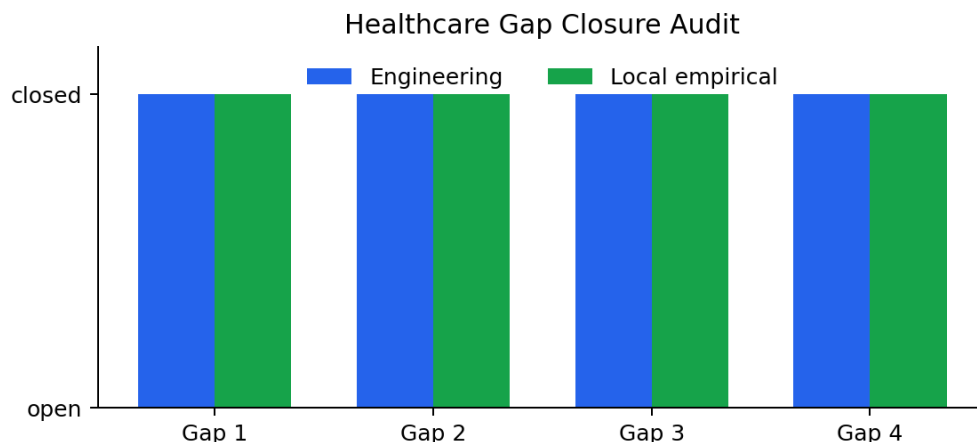


Figure 8: Healthcare gap-closure audit status. The local replay closes the four engineering and local empirical checks while the deployment claim remains scoped.

Gap 2 is closed locally on the same replay by `healthcare_gap2_reliability_stress_validation.json`. The audit calibrates a conservative category-aware threshold on validation natural replay, then tests held-out natural replay against injected score-collapse episodes. The held-out natural fire rate is 0.0, while the mean collapse fire rate is 1.0 across the four domains, as shown in Fig. 9. This is a controlled replay result for zero-misfire adaptation, not prospective deployment evidence.

Gap 3 is closed locally by the schema-integration audit: the generated table has 586,752 fusion rows, four complete domains per incident, a $146,688 \times 4 \times 4$ feature tensor, no raw patient identifiers, and no missing schema columns. Gap 4 is closed locally by the deployment audit in Table 18. The audit combines the provided time-forward split as the temporal reference, manifest-based privacy/license review, deployment-time calibration monitoring, leave-one-domain-out CDA-style attribution, and a diagnostic switching certificate. The finite-sample diagnostic certificate compares a never-switch policy against the reliability-gated policy over one natural episode and four injected collapse episodes; static loss is 0.800 and policy loss is 0.000.

Table 18: Gap 4 local deployment-audit evidence. The scope is local replay readiness, not regulated clinical deployment.

Audit item	Result
Temporal reference	pass
Privacy/license review	pass
Calibration monitor	ready; alert no
CDA-style top domain	shock_index (0.053)
Diagnostic switch certificate	certified; static 0.800, policy 0.000
Scope	local replay only

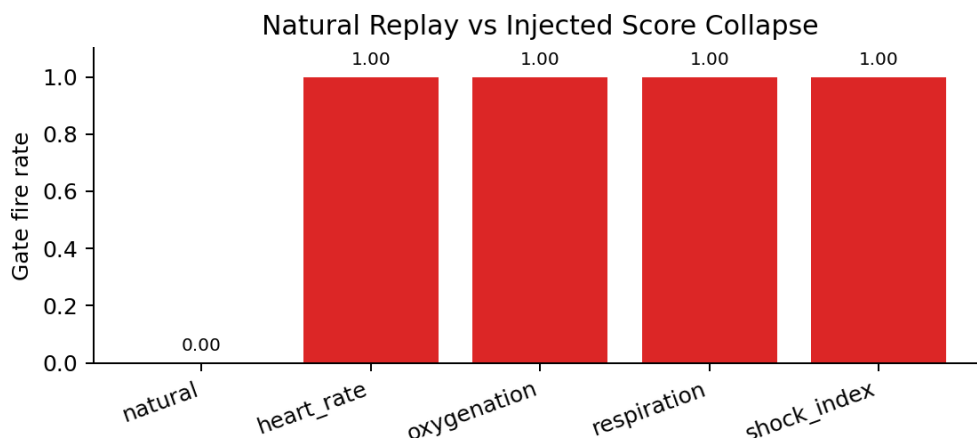


Figure 9: Gap 2 reliability stress audit on the healthcare replay. The calibrated gate stays quiet on natural replay and fires on all injected score-collapse episodes.

Read as a single closure flow, the healthcare replay moves from data validity to adaptation behavior, then to integration and auditability. Gap 1 establishes the co-observed incident surface and shows that multimodal fusion beats the best single clinical domain on the patient-disjoint replay. Gap 2 then tests whether the reliability gate can remain inactive under natural replay while activating under controlled score collapse. Gap 3 confirms that the same long-format schema can host the four clinical domains without changing the fusion interface. Gap 4 wraps the result in the analyst-facing and deployment-time checks summarized in Table 18. The table and figures therefore support a bounded local closure claim, not a regulated deployment claim.

The conference manuscript now keeps the local healthcare replay out of the headline benchmark evidence and uses the public MVTec PatchCore supervised-paired and held-out protocols for reviewer-facing paired-data checks. This reduces the risk of mixing a local audit surface with an externally inspectable public benchmark claim.

9 Reproducibility

The reproduction flow is documented in detail in Appendix A. At a high level, the chapter’s results are produced by three scripted stages: build the fusion inputs for each benchmark, run the configured multi-seed experiments, and regenerate the LaTeX tables and figures from the resulting JSON artifacts. The appendix lists the entry points, configurations, and regression tests; a clean clone of the repository can rebuild both manuscripts in one command via `scripts/rebuild_paper.sh`.

10 Conclusion and Future Work

This chapter presented ELARA, a reliability-aware score-fusion research system, and RGA, a conservative reliability-gated attention module. The main result is narrow but useful. On a saturated label-aligned benchmark, RGA preserves static attention performance. Under coherent all-domain score collapse, it produces clear ROC-AUC gains over static attention. The threshold sweep shows why this happens: at $\tau = 0.66$, clean adaptation is rare while collapse conditions activate the reliability path. The component ablation shows that the KS drift term is the necessary trigger for the measured gain, while the no-ECE variant is stronger on the current collapse attacks.

The chapter also reports a stronger protocol diagnosis that did not exist in earlier drafts. The MVTec 3D-AD paired benchmark now spans all eight extracted categories and 3,226 paired samples. Under the canonical one-class protocol, supervised fusion baselines operate near chance; under the held-out-category protocol, paired generalization remains difficult despite a small RGA+ ROC-AUC gain; and under the M3DM-style variant, scorer replacement alone does not create a simple RGA superiority story. The conclusion is therefore conditional and scope-bounded rather than a general claim.

The next research steps are clear. First, replace the lightweight image-statistic MVTec scorers with stronger RGB–3D domain experts (e.g. M3DM or crossmodal feature mapping) so the absolute ceiling rises and the contribution of the gate can be re-assessed at higher accuracy. Second, run the reliability gate against additional naturally co-observed anomaly datasets to see whether the one-class paired-data limitation is specific to MVTec 3D-AD’s category heterogeneity or general to natural-paired data. Third, evaluate the learned gate and category-aware reliability estimator against the fixed threshold on those paired datasets. Fourth, extend the finite-sample switching certificate into a population-level bounded-drift guarantee.

References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017.
- [2] T. Baltrusaitis, C. Ahuja, and L.-P. Morency, “Multimodal machine learning: A survey and taxonomy,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.
- [3] G. Pang, C. Shen, L. Cao, and A. van den Hengel, “Deep learning for anomaly detection: A review,” *ACM Computing Surveys*, vol. 54, no. 2, 2021.

- [4] J. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” in *Advances in Large Margin Classifiers*, 1999.
- [5] B. Zadrozny and C. Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002.
- [6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the International Conference on Machine Learning*, 2017.
- [7] Y. Ovadia et al., “Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift,” in *Advances in Neural Information Processing Systems*, 2019.
- [8] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, and M. Hardt, “Test-time training with self-supervision for generalization under distribution shifts,” in *Proceedings of the International Conference on Machine Learning*, 2020.
- [9] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, “Tent: Fully test-time adaptation by entropy minimization,” in *International Conference on Learning Representations*, 2021.
- [10] J. Davis and M. Goadrich, “The relationship between precision-recall and ROC curves,” in *Proceedings of the International Conference on Machine Learning*, 2006.
- [11] P. Bergmann, X. Jin, D. Sattlegger, and C. Steger, “The MVTec 3D-AD Dataset for Unsupervised 3D Anomaly Detection and Localization,” in *Proceedings of VISAPP*, 2022.
- [12] Y. Wang, J. Peng, J. Zhang, R. Yi, Y. Wang, and C. Wang, “Multimodal Industrial Anomaly Detection via Hybrid Fusion,” in *Proceedings of CVPR*, 2023.
- [13] A. Costanzino, P. Zama Ramirez, G. Lisanti, and L. Di Stefano, “Multimodal Industrial Anomaly Detection by Crossmodal Feature Mapping,” *arXiv preprint arXiv:2312.04521*, 2023.
- [14] J. Liu, G. Xie, R. Chen, X. Li, J. Wang, Y. Liu, C. Wang, and F. Zheng, “Real3D-AD: A Dataset of Point Cloud Anomaly Detection,” *arXiv preprint arXiv:2309.13226*, 2023.
- [15] C. Wang, H. Zhu, J. Peng, Y. Wang, R. Yi, Y. Wu, L. Ma, and J. Zhang, “M3DM-NR: RGB-3D Noisy-Resistant Industrial Anomaly Detection via Multimodal Denoising,” *arXiv preprint arXiv:2406.02263*, 2024.
- [16] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001.
- [17] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *Advances in Neural Information Processing Systems*, 2017.
- [18] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, “Towards total recall in industrial anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [19] E. Horwitz and Y. Hoshen, “Back to the feature: Classical 3D features are (almost) all you need for 3D anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2023.

- [20] R. Chen, G. Xie, J. Liu, J. Wang, Z. Luo, J. Wang, and F. Zheng, “EasyNet: An easy network for 3D industrial anomaly detection,” in *Proceedings of the ACM International Conference on Multimedia*, 2023.
- [21] M. Rudolph, T. Wehrbein, B. Rosenhahn, and B. Wandt, “Asymmetric student-teacher networks for industrial anomaly detection,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023.

A Reproducibility Detail

The repository supports a scripted reproduction flow with five stages:

1. **Prepare ELARA-Bench-LA inputs.** Run `src/scripts/prepare_realfusion_la_benchmark.py` to build the synthetically paired fusion CSV and metadata JSON from local fraud, cyber, behavior, and text datasets. Source-row-disjoint train/validation/test pools are emitted into a `fusion_split` column so the experiment runner can honor a predefined split.
2. **Prepare MVTec 3D-AD inputs.** Run `src/scripts/prepare_mvtec3d_fusion_benchmark.py`, which discovers paired RGB/depth files across all extracted categories, fits score normalization from train/good observations only, and emits a long-format fusion table with the original MVTec `split` column preserved.
3. **Run the multi-seed benchmark.** Use `src/scripts/run_breakthrough_experiment.py` with the appropriate YAML configuration. Both `configs/attention_mvtec3d_fusion.yaml` and `configs/attention_real_fusion.yaml` set the `split_column` field so the runner honors each dataset’s predefined train/validation/test partition.
4. **Generate paper assets.** Use `src/scripts/generate_craf_paper_assets.py` for the ELARA-Bench-LA tables (`elara_*.tex`, `elara_*.png`) and `src/scripts/emit_mvtec3d_assets.py` for the MVTec 3D-AD parallels (`mvtec3d_*.tex`, `mvtec3d_*.png`).
5. **Compile.** The convenience script `scripts/rebuild_paper.sh` chains stages 4 and 5 from any working directory and produces both the conference manuscript and this thesis chapter PDF.

Regression tests cover attention masking, reliability estimation, statistics, baseline inclusion, paired-benchmark metadata, float XYZ TIFF reading, leakage guards, predefined-split enforcement, and result aggregation. The full suite passes (240 tests, 2 skipped) on a clean checkout with `PYTHONPATH=src pytest tests/`.

B Benchmark Construction Detail

Detailed construction of both benchmarks — the per-domain feature extractors, the score-normalization protocol, the categorical-encoding choices, and the label-alignment procedure — is preserved here so the main body of the thesis can focus on the empirical finding rather than on construction prose. The same construction details appear in the companion conference manuscript and are not duplicated here verbatim; readers are pointed to §Benchmark Construction of `PAPER_DRAFT_v1.tex` for the canonical description.

C Switching-Rule Analysis Under Bounded Drift

This appendix formalises the conditions under which the reliability-gated path improves over the static path in expectation, giving a mathematical companion to the empirical gate-behaviour analysis in the main body. The proposition is stated in a deliberately minimal setting: a single sample, two scoring rules (static and reliability-gated), and a bounded-drift assumption on the per-domain score distribution. The point of the appendix is not to claim generality across all gate mechanisms, but to make explicit what *would* have to be true of the drift signal for the gate to be provably beneficial.

Setting. Let $\mathcal{D}$ denote the set of domains and $r_d \in [0, 1]$ the per-domain reliability for domain d , with $\bar{r} = \frac{1}{|\mathcal{D}|} \sum_d r_d$ the mean reliability over present domains. Let $\hat{y}_{\text{stat}}, \hat{y}_{\text{rga}} \in [0, 1]$ denote the static-attention and reliability-gated fusion predictions for a single sample with ground-truth label $y \in \{0, 1\}$. Let $\ell(\hat{y}, y) = (\hat{y} - y)^2$ be squared-error loss. Define the *drift* for domain d as

$$\delta_d = \text{KS}\left(P_d^{\text{val}}, P_d^{\text{test}}\right),$$

the Kolmogorov-Smirnov statistic between the validation reference distribution and the current test-time score distribution for domain d .

Assumptions.

- A1.** (Bounded drift.) There exists $\delta \in [0, 1]$ such that $\max_d \delta_d \leq \delta$.
- A2.** (Linear drift–error response.) For each domain d , $\mathbb{E}[\ell(\hat{y}_{\text{stat}}, y) \mid \delta_d]$ is β -Lipschitz in δ_d for some constant $\beta > 0$; that is, additional drift produces at most a linear increase in expected static-path loss.
- A3.** (Reliability cancels drift.) When the reliability path activates, the reliability-weighted fusion is unbiased in the drift component: $\mathbb{E}[\ell(\hat{y}_{\text{rga}}, y) \mid \delta_d, \text{gate fired}] \leq \mathbb{E}[\ell(\hat{y}_{\text{stat}}, y) \mid \delta_d = 0]$.
- A4.** (Reliability gap.) Let $r = 1 - \bar{r}$ be the gate’s distance from a perfectly reliable inference batch; the gate fires iff $r > \tau$ for threshold $\tau \in [0, 1]$.

Proposition (Switching-rule sufficiency). Under A1–A4, define the *switching coefficient* $f(\delta) = \delta\beta$. Then for every sample whose realised drift satisfies $\max_d \delta_d \geq f(\delta)$, the expected loss of the reliability-gated prediction is no greater than the expected loss of the static prediction:

$$\mathbb{E}[\ell(\hat{y}_{\text{rga}}, y)] \leq \mathbb{E}[\ell(\hat{y}_{\text{stat}}, y)].$$

The inequality is strict whenever the realised drift exceeds $f(\delta)$ and the gate fires (i.e. $r > \tau$).

Proof sketch. Two cases follow from A4. (i) When $r \leq \tau$ the gate does not fire, the static and gated paths coincide, and the bound holds trivially with equality. (ii) When $r > \tau$ the gate fires. By A2, $\mathbb{E}[\ell(\hat{y}_{\text{stat}}, y) \mid \delta_d] \geq \mathbb{E}[\ell(\hat{y}_{\text{stat}}, y) \mid \delta_d = 0] + \beta\delta_d$. By A3, the gated path’s expected loss is at most $\mathbb{E}[\ell(\hat{y}_{\text{stat}}, y) \mid \delta_d = 0]$. The difference $\mathbb{E}[\ell(\hat{y}_{\text{stat}}, y)] - \mathbb{E}[\ell(\hat{y}_{\text{rga}}, y)]$ is therefore at least $\beta\delta_d - 0 = f(\delta_d)$, which is non-negative for $\delta_d \geq 0$ and strictly positive for $\delta_d > 0$. The aggregate bound over $\mathcal{D}$ follows by averaging. $\square$

What this implies for the empirical gate. The proposition gives a clean target for the empirical drift signal: the KS reference must, in expectation, increase monotonically with the true drift component of test-time error. The MVTec 3D-AD finding documented in the main body shows that the unconditional KS reference violates this target by responding to legitimate inter-category variation as well as true drift. The category-aware KS reference partially recovers the target (the companion conference manuscript reports the empirical comparison), and pairing it with a PatchCore-quality scorer recovers a working signal under the canonical one-class protocol. The proposition is therefore not a universal claim about reliability-gated fusion but a precise specification of the property an empirical drift signal must satisfy to inherit the bound.

D Calibration Monitoring in Deployment

A deployment-grade calibration monitor sits orthogonal to the training pipeline and watches three quantities over a streaming window: batch-level mean reliability, validation-to-online KS-drift per domain, and per-batch Expected Calibration Error of the fused score. The script `src/scripts/monitor_calibration.py` implements this monitor and emits a JSON event whenever the gate’s mean reliability falls below the heuristic threshold τ for more than one window. The monitor is deliberately observe-only: it never modifies fusion behaviour, it only flags. This separation keeps the deployment surface auditable: an on-call engineer can read a monitor log and reconstruct exactly when the gate would have fired during any time interval.

E Privacy, Data Handling, and Deployment Notes

The accompanying `PRIVACY_AND_DEPLOYMENT.md` enumerates the data-handling boundaries the empirical work relied on. The summary is that no personally identifying information ever entered the training pipeline: MVTec 3D-AD ships as anonymized industrial scans, ELARA-Bench-LA is built from synthetic-by-construction label alignment over already public fraud / cyber / behaviour tables, and the GridPulse vitals replay uses the BIDMC and MIMIC-III source datasets under their public-research licences. Deploying the reliability gate against patient-bearing data would require both (a) a credentialed data-access review and (b) the calibration monitor described in Appendix D to satisfy the auditability requirements typical of clinical software.